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DEVELOPMENT ARTICLE

Enhancing the flipped classroom model with generative AI and Metaverse technologies: insights from lag sequential and epistemic network analysis

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Abstract
The Flipped Classroom Model (FCM) has gained widespread acceptance in higher education as an effective pedagogical strategy. Despite its success, the FCM still faces persistent concerns, including a lack of personalized interaction, limited application to introductory courses, and insufficient analysis of the learning process. The integration of generative artificial intelligence (AI) with the Metaverse may significantly enrich and revitalize FCM by introducing affordances that support personalized and immersive learning experiences. This study examined the potential of FCM enhanced by an AI-Generated Metaverse (AIGM) to improve learning outcomes in higher education. Through introducing an AI-powered virtual tutor, this study aimed to address existing concerns and shed light on the nature of knowledge-building interactions between students and the AI tutor. The study involved 94 university graduates who participated in an experimental FCM session. Data on students' learning performance were collected through pre/posttests, pre-class video engagement, and analyzed dialogic interactions between students and the AI virtual tutor. The findings indicated that the AIGM-FCM positively influenced learners' academic performance. Interaction patterns revealed that students predominantly engaged with the AI virtual tutor at the onset and at the end of their learning sessions, primarily for the purpose of asking questions. However, the educational dialogues suggested that the depth of collaborative knowledge construction between students and the AI tutor remained at a basic level. Notably, students with lower academic performance were more likely to initiate inquiry-based dialogues with the AI tutor compared to their higher-achieving peers exhibiting a more focusing-oriented approach. In light of these discussions, the implications of this study are also deliberated.

Keywords Generative artificial intelligence · Metaverse · Flipped classroom model · Higher education

Extended author information available on the last page of the article

Introduction

The Flipped Classroom Model (FCM) has been acknowledged as an effective instructional strategy in higher education (Akçayır & Akçayır, 2018; Al-Samarraie et al., 2020; Wang et al., 2023a). Despite its proven efficacy, FCM still faces significant challenges, such as a lack of support for providing timely and personalized feedback to students during their independent pre-class work. This limitation often confines the use of FCM to introductory-level courses (Al-Samarraie et al., 2020) and can negatively impact student engagement and the overall effectiveness of the model (Akçayır & Akçayır, 2018). Furthermore, previous assessment studies on flipped learning have not adequately addressed learners' behavior since the self-guided pre-class activities—which primarily consisted of watching videos—lack robust technical means for tracking or analyzing students' engagement with the learning materials (Cheng et al., 2022a). Although learning analytics could monitor and analyze learners' actions within digital environments, their applications in FCM research are still limited (Kang et al., 2024). The inadequacy of other pedagogical improvements for augmenting FCM's efficacy, coupled with the lack of seamless technology integration, hinders researchers' ability to attain comprehensive understanding to further develop the flipped learning methodologies (Bredow et al., 2021; Cheng et al., 2022a).

In recent years, the educational landscape has undergone a transformation with the rapid advancement of Artificial Intelligence (AI) and the Metaverse (Bobko et al., 2024; Hwang & Chien, 2024; Wang et al., 2022; Zhai et al., 2021), leading to innovations such as enhanced feedback mechanisms, reasoning facilitation, and adaptive learning (Zhai et al., 2021). Furthermore, the rise of generative AI and the Metaverse has introduced new learning modalities, further enriching students' educational experiences (Hwang & Chen, 2023; Wang et al., 2022). According to the AI-Generated Metaverse (AIGM) framework (Lee et al., 2023), generative AI and the Metaverse are anticipated to converge synergistically, with generative AI facilitating content creation and the Metaverse offering immersive experiences. Cutting-edge technologies based on AIGM framework can be integrated to support AI virtual tutors, which assist personalized instruction and foster deeper knowledge construction. Hence, AIGM presents significant opportunities to rejuvenate FCM (Scardamalia & Bereiter, 2022; Wu et al., 2023b).

Nonetheless, despite the promising potential of AIGM, there remains a need for more empirical evidence regarding its actual impact on FCM. Moreover, the changes and variations in learner FCM interaction patterns facilitated by these technologies are not yet well understood. To bridge these gaps, this study endeavors to examine the effectiveness of the Flipped Classroom Model enabled by generative AI and Metaverse technologies (AIGM-FCM). In higher education settings. It also scrutinizes the collaborative knowledge-building patterns among learners of different performance levels within the FCM facilitated by AI virtual agents. Employing integrated learning analytics, this research intended to shed light on interactions and dialogues with these virtual agents, thereby enriching our comprehension of their contribution to the flipped learning paradigm. The investigation was driven by the following research questions:

RQ1 What are the impacts of AIGM-FCM on learners' academic performances?
RQ2 What are the differences in the behavioral sequences of high-performance and low-performance students within the AIGM-FCM?

RQ3 What are the differences in the collaborative knowledge building patterns of high-performance and low-performance students during their interaction with the AI virtual tutor within a flipped classroom?

Literature review

The application of FCM in higher education and its challenges

The FCM embodies a suite of pedagogical strategies that prompt students to complete preparatory work before class, thereby enabling deeper engagement with in-class activities (Birgili et al., 2021). This model reverses the conventional sequence of lectures followed by homework, redefining pre-class preparation and in-class interaction (Brewer & Movahedazarhouligh, 2018). It enhances classroom learning by transferring the acquisition of foundational knowledge to pre-class time, which in turn allows for more efficient use of in-class periods, and fosters the development of learners' twentyfirst-century skills (Akçayır & Akçayır, 2018). As a learner-centric instructional method, FCM necessitates a higher degree of self-regulation from students in their pre-class studies. Through reviewing the implementation, efficacy, and quality of flipped learning in higher education, Brewer and Movahedazarhouligh (2018) identified student autonomy and proactivity as pivotal features of the FCM and emphasized the responsibility of learners to independently navigate pre-class materials, thereby constructing a robust foundation of knowledge.

Extensive empirical studies have underscored the effectiveness of the FCM in terms of enhancing academic outcomes such as performance (Akçayır & Akçayır, 2018), motivation (Challob, 2021), and higher-order thinking skills (Bredow et al., 2021) in higher education; yet, its implementation is not without challenges and has been met with skepticism and various obstacles. First, pre-class learning that lacks sufficient interactivity and guidance can result in the absence of timely feedback and personalized instruction focused on individuals' needs, potentially leading to an excessive cognitive burden for learners (Akçayır & Akçayır, 2018; Zhai et al., 2017). Second, the conventional FCM may be more effectively applied to courses that focus on introductory instruction, yet it calls for significant modifications to accommodate courses necessitating experiential learning where more inquiry-oriented knowledge building is essential (Al-Samarraie et al., 2020). Third, the assessment of FCM has predominantly concentrated on grades and self-reported data, and there is a paucity of objective metrics that capture student engagement and learning strategy utilization within the flipped learning environment, particularly in those enhanced by technology (Cheng et al., 2022a).

Although some researchers have attempted to enhance the effectiveness of FCM by incorporating more pedagogical strategies, Bredow et al. (2021) found in their meta-analysis of 317 independent studies that course design had only a modest impact on the effectiveness of flipped learning. According to the meta-analysis results, effect sizes (Hedges' g) for all assignment and quiz settings, whether pre-class or in-class, were below 0.5. Additionally, varying time allocations in flipped learning showed no significant effect on overall effectiveness. Furthermore, the teachers' workload associated with providing personalized feedback, as well as the observation bias of self-reported data, both underscore the importance of technology-enhanced instructions and learning analytics in the online flipped learning environment (Hwang et al., 2021; Lee & Choi, 2019).

The integration of generative AI with the Metaverse in education

Generative AI and the Metaverse are emerging as transformative technologies with profound potential to revolutionize higher education (Hakiki et al., 2023; Wang et al., 2022). Generative AI models, including ChatGPT, Gemini, and Sora, leverage vast datasets to produce content and engage in personalized dialogues with learners. This model offers dynamic responses rather than static, rule-based interactions (Baidoo-Anu & Ansah, 2023; Sætra, 2023). They are capable of creating instructional material and adapting conversations in real time to address individual student queries (Dwivedi et al., 2023). The Metaverse, a collective of 3D virtual environments, facilitates immersive interactions and simulates real-world activities, while also capturing comprehensive behavioral data from learners (Wang et al., 2022; Zhai et al., 2023).

The synergy between generative AI and the Metaverse is not merely coincidental but is characterized by their complementary functionalities (Lee et al., 2023). Lee et al. (2023) envisioned an AIGM framework where the integration of generative AI with the Metaverse creates a cohesive ecosystem, highlighting the pivotal role of AI in content generation and the Metaverse in providing engaging, immersive experiences. Generative AI enriches the interactivity and its embodiment of virtual agents in Metaverse environments, allowing virtual tutors to effectively support students by conducting learning analysis and facilitating personalized knowledge acquisition (Qadir, 2023). Moreover, the Metaverse optimizes the interface for interactive, collaborative educational endeavors with generative AI. The synthesis offers a unified digital space where the learner's dialogue and interaction with AI are not merely facilitated but also seamlessly documented and analyzed for continuous improvement (Topsakal & Topsakal, 2022).

This convergence is particularly promising for enhancing the FCM (see Fig. 1), where the interactivity of the Metaverse and the adaptability of generative AI are poised to create fertile ground for student-centered learning (Wu et al., 2023b). The Metaverse platforms

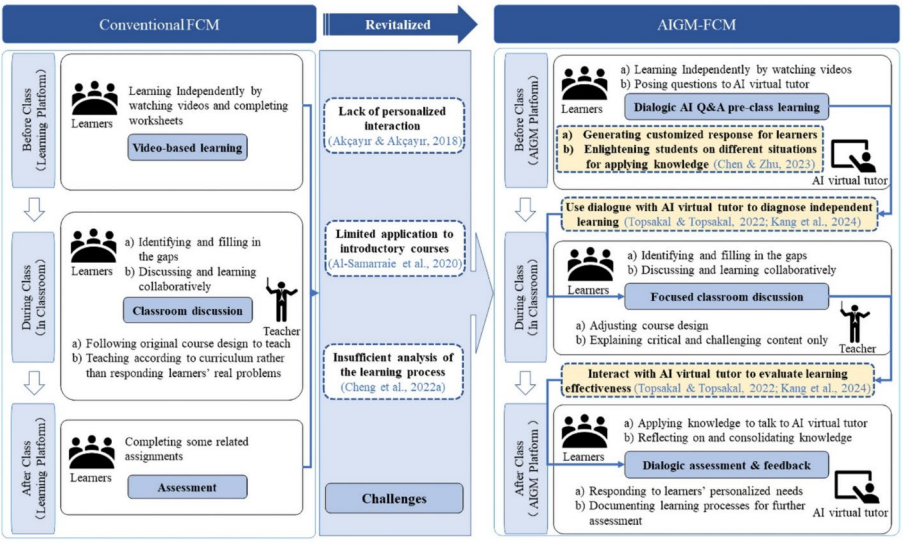


Fig. 1 Revitalized AIGM-FCM compared with conventional FCM

are adept at forging immersive, interactive spaces where AI-driven virtual tutors offer customized support, spanning the spectrum from pre-class preparation to real-time progress assessments, and applying teaching strategies that are finely tuned to the needs of individual learners (Han et al., 2023). Moreover, the capacity of generative AI agents to produce new content for learners through natural language dialogue aligns with the primary principles of knowledge building, specifically "Idea Improvements" and "Knowledge-Creating Dialogue" (Scardamalia & Bereiter, 2022). This further supports information retrieval and knowledge construction, and widens students' perspectives on different situations for knowledge application (Chen et al., 2023). In addition, technology-enhanced environments offer the opportunity to trace individual learners' actions and their collaboration with AI tutors to provide more data-driven analysis and intervention (Kang et al., 2024). For instance, Kang et al. (2024) revealed the formation of cognitive networks during students' learning process by collecting their multimodal interaction data from immersive environments, which provided deeper insights into the productive knowledge co-construction.

In the AIGM-FCM, the virtual tutor and human teacher collaborate to support students throughout the entire learning process. Before the face-to-face lessons, students are briefed on the limitations of large language models (LLMs), such as inaccuracy and bias. They are then instructed to treat the outputs of LLMs as knowledge resources rather than facts, and to build their understanding of the topics through interactions and dialogues with the AI tutor (Chen et al., 2023). Although the generative AI supported by LLMs have certain constrains, the purpose of interacting with the AI tutor is not to provide learners with perfect answers, but to serve as epistemic resources that provoke students' thinking (Salinas-Navarro et al., 2024). These pre-class dialogues also assist the human teacher in diagnosing students' preparation levels and allow them to focus on more critical and innovative issues during class, thereby improving classroom efficiency (Topsakal & Topsakal, 2022). Additionally, the human teacher's instructions help prevent confusion that might arise from inaccuracies and biases inherent in LLMs and ensure the responsible and ethical use of LLMs (Salinas-Navarro et al., 2024).

Despite the apparent potential, empirical research on the FCM augmented by generative AI within the Metaverse remains sparse (Dimitriadou & Lanitis, 2023). The intricacies of the mechanisms underlying their effectiveness in educational contexts are yet to be fully unraveled and so warrant further investigation (Hwang & Chien, 2022).

Investigation of the knowledge-building process in AIGM-FCM

Learning analytics are effective approaches to advance the understanding of students' learning behaviors and intrinsic learning mechanisms (Kang et al., 2024; Yang et al., 2015). Lag Sequential Analysis (LSA) and Epistemic Network Analysis (ENA) are two prevailing learning analytic techniques that have popularized the investigation of students' behavioral sequences and have provided insights into their thinking processes. For instance, Yang et al. (2015) adopted LSA to compare the programming learning behavioral patterns between control and experimental groups. This study further investigated learning processes affected by teaching strategies beyond test results. ENA serves as another useful technique to unveil learners' paradigms of knowledge building and conception understanding (Chang & Tsai, 2023; Hod et al., 2020).

Using learning analytics to investigate the process of collaborative learning in immersive and AI-supported learning environments has garnered more attention in the past few years. For instance, Zhan et al. (2022) employed LSA to explore collaborative problem-solving

activities among learners in an immersive laboratory. This analysis revealed students' collaborative pathways and provided implications for the design of future collaborative activities. Meanwhile, Wang et al. (2023b) investigated human-AI interactions in language learning using ENA. The ENA results enabled the researchers to further examine and perform pairwise comparison of different students' learning experiences with AI. Furthermore, researchers also incorporate these two methods because ENA extends the scope of LSA from external behaviors to internal epistemic processes. In addition, LSA can position students' cognitive interaction in the entire collaborative activity process to assist more comprehensive discernment (An & Zhang, 2024; Tan et al., 2022). These findings illustrate the potential use of LSA and ENA in analyzing the learning process within the AIGM framework, leading to valuable insights into the effects and mechanisms of utilizing AIGM-FCM in higher education.

In conclusion, the AIGM framework presents the significant potential for revitalizing the FCM with personalized guidance, improved knowledge acquisition, and comprehensive tracking of learners' learning journeys. Nevertheless, the current lack of empirical evidence for the effectiveness of AIGM-FCM in higher education underscores the necessity of this study which sought to investigate the impact of AIGM-FCM on students' learning outcomes in higher education. Moreover, the dynamics of human-AI knowledge building within the FCM also warrant further investigation by employing LSA and ENA.

Methodology

Participants and educational context

This study involved 94 postgraduate students from a university in eastern China, majoring in Educational Management, Chinese Education, and Modern Educational Technology (see Table 1). Each participant had teaching experience and aimed to integrate their learning into practical teaching scenarios. The study utilized a specially developed learning platform, *MyEduChat* (see Fig. 2), to deliver an AIGM-FCM environment. This platform featured instructional videos, reading materials, and a virtual agent tutor equipped with OpenAI ChatGPT-3.5, facilitating AI-generated questions and answers for effective human-AI interaction and knowledge building. Furthermore, Text-To-Speech (TTS) technologies were used to vocalize the tutor's responses, enhancing the human-AI interaction with 3D agent images, expressions, and gestures for more engaging visual cues (Chen

Table 1 Demographic profile of participants ($N=94$) (Gender is categorized in the context of Chinese culture and norms)

Variables	Categories	*n* (%)
Gender	Male	8 (8.51%)
	Female	86 (91.49%)
Age	< 25	16 (17.02%)
	25–30	57 (60.64%)
	30–35	16 (17.02%)
	> 35	5 (5.32%)
Major	Educational Management	48 (51.06%)
	Chinese Education	41 (43.62%)
	Modern Educational Technology	5 (5.32%)

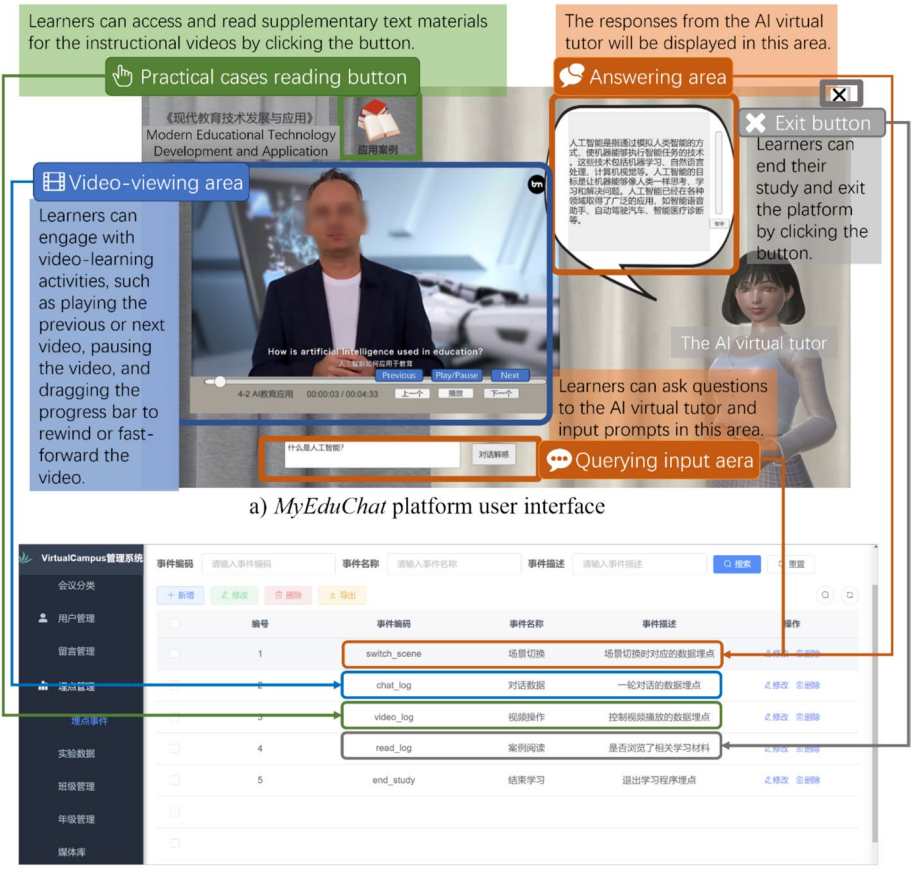


Fig. 2 User interface and backend management system of the *MyEduChat* platform

et al., 2019). To track and analyze interactive learning behaviors, students were required to register individual accounts on *MyEduChat*, enabling the documentation of video viewing data, material reading logs and dialogues with the AI virtual tutor in real time.

Pedagogical design

The FCM was integrated into the "Modern Educational Technology Development and Application" course using the *MyEduChat* platform in this study. Unlike traditional humanities or social science courses, "Modern Educational Technology Development and Application" also requires integrating technology into educational settings. The course entails the application of artificial intelligence tools to tackle practical teaching problems within the Technological Pedagogical Content Knowledge (TPACK) framework, along with the design and coding of instructional programs. It necessitates a higher level of cognition to construct knowledge structures rather than merely understanding the content. The course materials were divided into four modules, each including three instructional videos. The

entire experimental course, consisting of eight class sessions, was conducted from July 1 to 13, 2023.

Students were informed that the AI virtual tutor on the platform could engage in dialogues as part of the FCM course design, assisting them in understanding the course content. The approach aligns with the emphasis on applying guiding principles for knowledge building rather than relying on prescribed procedures (Scardamalia & Bereiter, 2022). This study primarily followed the FCM framework in designing instructional activities, incorporating principles that facilitated students' construction of TPACK. Six intertwined principles were included, and the AIGM-FCM provided corresponding support for learners' knowledge construction, as detailed in Table 2.

Guided by these principles, the collaborative flipped learning between learners and the AI virtual tutor unfolded through specific procedures. Prior to class, students were required to engage in autonomous learning of the course material, which included video lectures, reading instructional materials and posing questions to the AI virtual tutor. The AI virtual tutor was designed to refine the students' ideas and to foster a knowledge-building

Table 2 Knowledge building principles and the AIGM-FCM supports

Knowledge building principles	AIGM-FCM supports
• *Real Ideas and Authentic Problems* Knowledge-building problems are closely related to real-world contexts around learners, striving to clarify issues that help them understand the world	• In the AIGM-FCM learning environment, learners engage with content focused on the application of modern educational technologies to teaching practices, closely aligning with the actual development needs of learners
• *Epistemic Agency* Learners partially assume the role of teachers, managing their own learning goals and plans. In the process of knowledge building, they promote knowledge advancement through contrasting or challenging ideas, rather than relying on others to chart the course	• The AIGM-FCM mandates students to independently organize their learning prior to class without direct teacher guidance; similarly, the AI virtual tutor encourages learners to exercise their agency by initiating prompts for collaborative interaction and construction
• *Improvable Ideas* Learners work continuously to enhance the quality, coherence, and utility of their ideas	• The AI virtual tutor is designed to facilitate knowledge building beyond text and video resources within the course, enabling learners to continuously refine these ideas. With the limitations of LLMs explained, students are encouraged to critically evaluate their interactions with generative AI and further improve their instructional designs
• *Knowledge Building Discourse* Learners do not merely propose ideas but actively engage in discursive practices, integrating ideas in dialogical processes to advance knowledge	• The affordances of the AI virtual tutor's multi-turn responses support dialogic practices with learners. The pre-class learning records on the AIGM-FCM platform offer teachers insights into students' independent learning, enhancing efficient classroom interactions through feedback and guidance on unresolved obstacles
• *Pervasive Knowledge Building* Knowledge building is not limited to specific occasions or subjects, but occurs across contexts	• Learners come from diverse disciplines, and the AI virtual tutor, equipped with knowledge across these fields, assists learners in building knowledge on the application of modern educational technologies across various subjects
• *Embedded and Transformative Assessment* Assessment is an integral part of knowledge building, aiding knowledge advancement through reflection on knowledge gaps as work progresses	• Within the AIGM-FCM learning platform, reflective questions are provided to support learners in assessing their mastery and construction of knowledge

dialogue. Then, in-class sessions provided an opportunity to deeply discuss any knowledge gaps not resolved pre-class and to critically evaluate the responses from generative AI. Teachers could also evaluate pre-class knowledge construction based on the dialogue data between students and the AI virtual tutor, allowing for tailored course design focused on complex and critical knowledge points. Teacher instruction plays a critical role in the AIGM-FCM framework, as LLM-generated content is not always valid or unbiased. In this study, two educational technology experts evaluated the LLM-generated responses according to the evaluation criteria outlined by van der Lee et al. (2021). The results revealed the following ratings for the AI tutor's responses: informativeness (94.99%), relevance (90.23%), accuracy (95.78%), correctness (96.83%), and appropriateness (98.42%), with inter-rater reliability scores exceeding 0.722 across all dimensions. Therefore, it is essential for teachers to provide more accurate and detailed explanations in class and to correct any inappropriate content. After class, learners continued to refine their ideas based on their emerging understanding in dialogues with the AI virtual tutor, receiving assessments and feedback. An example of a theory concept map and TPACK-based instructional design created within AIGM-FCM is shown in Fig. 3.

Data collection and coding

This study utilized a single group pre/posttest research design, conducting academic performance assessments before and after the course. The test items were selected from a standardized question bank, and were tailored to match the specific instructional content while ensuring consistent difficulty levels. Examples include questions such as "What skills and competencies can be developed in project-based learning?" in the pretest and "What are some artificial intelligence algorithms? (List three)" in the posttest. These test items were approved after being reviewed by three scholars specializing in educational technology. At the end of this experimental course, valid pre- and posttest scores from a total of 88 participants were collected.

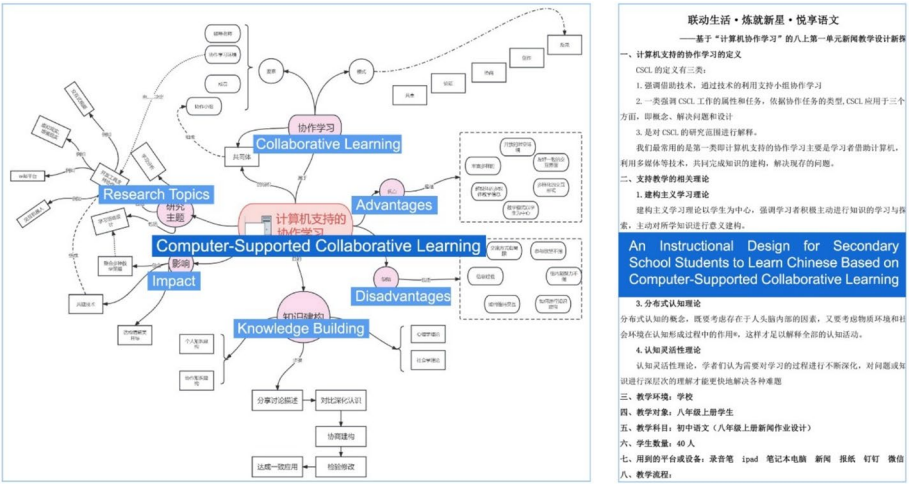


Fig. 3 Example of a theory concept map and TPACK-based instructional design

Building upon Wu et al. (2023a, 2023b) research on video-viewing engagement, this study annotated eight types of learning behaviors within the AIGM-FCM, as shown in Table 3. Learners accessed the platform multiple times during the course. To explore how students collaborated with the AI virtual tutor during flipped video-viewing learning, each series of video learning activities per log session was considered as a distinct learning session. Moreover, the study identified the end of a learning session by marking the benchmark learning activity using time stamps from the behavior log. A single learning session could encompass various behaviors, including video viewing (such as starting, rewinding, fast-forwarding, pausing, and continuing a video), reading materials, and posing questions to the AI virtual tutor. The platform automatically recorded all learner behaviors. Data cleaning was conducted to remove records of excessively repetitive clicks and obviously short learning duration. According to Zelenski et al. (2015), sessions shorter than 10 min (the median duration per module was 17.5 min) were excluded. As a result, 3906 valid learning behavior records were obtained.

To gain deeper insights into the collaborative knowledge-building process, this study further encoded the dialogue data between learners and the AI virtual tutor. Dialogue is considered an intersubjective interaction activity between individuals. According to Buber (1937), and with the expansion of the dialogue space, when combined with appropriate supporting pedagogy, constructive dialogue can also occur between educational technology and learners (Wegerif & Major, 2019). For the analysis of dialogues within knowledge-building environments, Cacciamani et al. (2018) developed the Conversational Functions for Knowledge Building (CF4KB) coding scheme, which includes coding for learners' dialogical actions such as inquiry, explanation, synthesis, evaluation, and reflection, among others. Regarding the analysis of technology-mediated dialogues, Hennessy (2023) proposed the Tech-SEDA framework, developed from a wide range of practical research examples. This framework provides researchers with a perspective for understanding how learners engage in dialogical forms such as building on ideas, focusing, reasoning, and challenging within technology-supported environments. Thus, we selected and adapted dialogue coding forms from the schemes of Cacciamani et al. (2018) and Hennessy (2023) to align with the objectives of the current study. These were then correlated with the most relevant knowledge-building principles (see Table 4).

After filtering out duplicate questions from learners during the waiting period for AI tutor responses, 688 valid dialogue instances were identified in their conversations. Among these valid instances, 327 were contributed by the learners and were subsequently coded. The coding process was independently carried out by two experts in educational technology. Inter-rater reliability was assessed by computing the Cohen's Kappa coefficient, revealing an agreement of 0.861 between the two raters. Any inconsistencies in codes were addressed through discussion between the two coders, leading to the determination of the final codes.

Data analysis

This study employed a quantitative research approach to examine the outcomes and processes of collaborative knowledge-building activities between learners and the AI virtual tutor within FCM. To address the first research question, we conducted paired-sample *t* tests on learners' pre- and post-experiment of knowledge transfer and retention test scores utilizing SPSS 26. A two-way analysis of variance (ANOVA) with repeated measures was employed as well to evaluate the effect of the AIGM-FCM on students' academic

Table 3 Learning behaviors recorded in the AIGM-FCM platform

Behavior	Codes	Description
Start a video	SV	Learners start playing an educational video
Rewind a video	RV	Learners drag the progress bar to replay the video
Fast-forward a video	FV	Learners drag the progress bar to fast forward the video
Pause a video	PV	While watching a segment of an educational video, learners press the pause button to temporarily stop the video playback
Continue a video	CV	While watching a segment of an educational video, learners resume playing the video after pausing
Read text materials	RM	During the learning process, learners open textual materials for study
Ask a question to the AI virtual tutor	AQ	During the learning process, learners ask questions to the virtual tutor supported by generative AI
End study	ES	Learners end this learning session and exit the platform

Table 4 Tech-SEDA for dialogue analysis between learners and the AI virtual tutor

Dialogue form	Codes	Description	Examples
Building on ideas	B	Request the AI tutor to build on, elaborate, or clarify some ideas	"How can the application of artificial intelligence in education improve teaching efficiency?"
Focusing	F	Guide or focus dialogue on a certain desired direction or towards certain key aspects	"Are there any case studies of AI applications in education?"
Challenge	CH	Question, doubt, fully/partially disagree, challenge or reject an idea	"The flipped classroom has certain advantages, but there are difficulties in practice. How should they be addressed?"
Reasoning	R	Explanation, justification, and comparison of ideas	"What is the relationship between complexity theory and systems theory?"
Inquiry	I	Pose probing questions and aim at authentic problem-solving	"How can classes with weaker foundations implement project-based learning?"
Co-ordination	CO	Collaboratively negotiate, synthesize ideas or cooperate with the AI tutor	"Some people say that artificial intelligence in education is similar to home tutors. Do you think this statement is reasonable?"
Reference Back	RB	Refer to previous knowledge, beliefs, teaching experiences or contributions	"Hello! How can we enhance the quality of online learning in the hybrid learning model? What strategies can we employ to prevent disengagement or lack of participation?"
Reference to Wider context	RW	Make links between what is being learned and a wider context	"How will AI evolve in the future?"
Reflection	RF	Evaluate or reflect on the dialogue process through metacognition	"The answer you generated doesn't solve my problem. Please improve it and add more professional expression."

performance across time. Prior knowledge level is a between-subjects factor, whereas test time is a within-subjects factor. Prior knowledge level included two levels (low, high), and test time consisted of two levels (pre-test, post-test).

Subsequently, learners' behavioral and dialogic data were segmented into higher- and lower-score groups based on their average performance in the posttest for a comparative analysis through LSA and ENA. The analysis of learners' behavior data through LSA was conducted using GSEQ 5.1. A chi-square test confirmed a significant relationship between the learning behaviors and their tallied frequencies, indicating that the most probable scenarios involved the repetition of similar activities. This suggests a tendency among users to engage in consistent actions. According to Bakeman and Gottman (1997), the computation of z-values assists in determining the significance of the sequences between two elements in LSA.

This study employed the ENA Web Tool (version 1.7.0) for analyzing coded dialogues between learners and the AI virtual tutor, and utilizing R for visual representations of the frequency differences in dialogue form. The ENA models the connections among codes by quantifying the co-occurrence of codes within conversations, generating a weighted network of these co-occurrences, alongside associated visualizations for each unit of analysis. The ENA algorithm utilizes a moving window approach to construct a network model for each line of data, illustrating the connections between current codes and those occurring previously (Siebert-Evenstone et al., 2017). We set the window length to 4, consistent with previous collaborative discourse analyses (Peeters et al., 2022; Zörgő et al., 2021).

The ENA model normalized networks for all analysis units before subjecting them to dimensional reduction, addressing potential variations in the number of coded lines across different units. Singular value decomposition was applied for dimensional reduction, producing orthogonal dimensions that maximize the variance explained by each dimension. For a more detailed explanation of the mathematics involved, refer to Shaffer et al. (2016). A Mann–Whitney test was conducted to test for differences in the location of points within the projected ENA space for units categorized as High and Low groups.

Results

Academic performance

To examine RQ1, a paired-samples t test was employed to analyze the data collected from assessments administered before and after the collaborative activities. The results, as presented in Table 5, show a significant difference in both the pretest and posttest ($MD = 0.129$, $t = 5.302$, $p < .01$, Hedges' $g = 0.775$), with the posttest scores statistically significantly higher than those of the pretest scores.

The learners' pretest scores were used to distinguish their prior knowledge levels. Table 5 summarizes the results of repeated measures ANOVA and indicates that students' prior knowledge level and test time had a significant interaction effect ($F = 29.463$, $p = .000$, $\eta_p^2 = 0.255$). According to the findings of the interaction effect (refer to Fig. 4), learners' academic performances were enhanced after completing the AIGM-FCM learning session. This improvement was observed regardless of their prior knowledge levels, with students who had lower prior knowledge even demonstrating more significant improvements (Table 6).

Table 5 The *t*-test results of academic performance

		Group	*N*	Mean	*S.D.*	*MD*	*t*
Academic performance		Pre-test	88	3.07	0.883	0.129	−5.302***
		Post-test	88	3.76	0.897		
Retention test	Pre-test		88	3.63	0.835	0.131	−4.694***
	Post-test		88	4.24	0.830		
Transfer test	Pre-test		88	2.52	1.406	0.174	−4.322***
	Post-test		88	3.27	1.111		

***$p < 0.001$

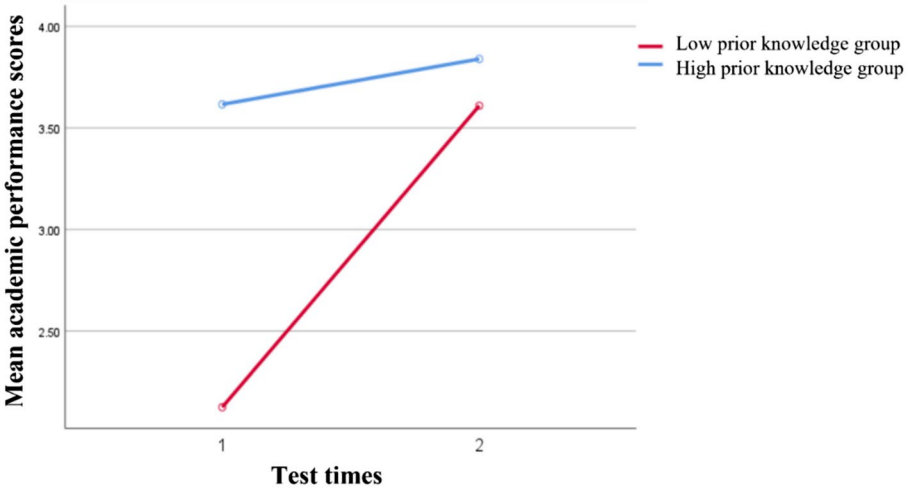


Fig. 4 Interaction of prior knowledge level and test time for academic performance

Table 6 Repeated measures ANOVA results of academic performance

Source	*SS*	*df*	*MS*	*F*	*p*	η_p^2
Prior knowledge level	31.664	1	31.664	63.613	.000	0.425
Error	42.807	86	0.498			
Test time	15.464	1	15.464	28.209	.000	0.247
Test time*prior knowledge level	16.151	1	16.151	29.463	.000	0.255
Error	47.144	86	0.548			

Adopting the grouping method delineated by previous studies (Chen & Wang, 2020; Cheng et al., 2024; Wu et al., 2020), individuals archiving posttest scores above the median (3.75) were classified into the high-achieving group (HG). Conversely, those with scores below this threshold were allocated to the low-achieving group (LG). This

median split approach resulted in 50 participants being categorized into the HG, with the remaining 38 assigned to the LG (see Table 7).

Learning behavior patterns from lag sequential analysis

To address the second research question, lag sequential analysis was applied separately to explore the learning behavior patterns of the high-performing group (HG) and the low-performing group (LG) learners. This analytical approach facilitated the examination of the significant behavioral sequences where one specific behavior was immediately followed by another. The frequencies of the eight identified codes, for both groups, are presented in Tables 8 and 9. The initial behavior is listed in the left-hand column, with the subsequent row extending to the right, indicating the frequency of users' next behavior type.

The analysis revealed that the volume of codes for the low scoring group (1863 codes) was less than that for the high scoring group (2043 codes). This discrepancy suggests that students in the low scoring group engaged in fewer learning activities compared to their counterparts in the high scoring group.

Table 7 Descriptive statistics for the posttest scores of the HG and the LG

Group	*N*	Overall post-test score		
		Range	Mean	*S.D.*
HG	50	4.0–5.0	4.41	0.36
LG	38	1.5–3.5	2.89	0.61

Table 8 Results of frequency transition in the HG

	SV	RV	FV	PV	CV	RM	AQ	ES	Totals
SV	526	34	45	93	5	22	145	66	936
RV	15	45	6	22	21	1	12	0	122
FV	41	6	74	7	2	0	5	7	142
PV	13	13	0	0	212	8	25	17	288
CV	50	16	7	152	4	2	19	7	257
RM	11	3	0	2	1	6	5	16	44
AQ	141	5	9	12	12	5	43	27	254
ES	0	0	0	0	0	0	0	0	0
Totals	797	122	141	288	257	44	254	140	2043

Table 9 Results of frequency transition in the LG

	SV	RV	FV	PV	CV	RM	AQ	ES	Totals
SV	395	19	29	126	12	15	69	42	707
RV	10	47	11	21	9	0	6	1	105
FV	23	17	81	8	7	0	2	3	141
PV	17	8	2	0	294	4	33	14	372
CV	75	10	16	208	4	2	12	9	336
RM	10	0	1	0	0	2	4	7	24
AQ	62	4	1	8	10	1	52	40	178
ES	0	0	0	0	0	0	0	0	0
Totals	592	105	141	371	336	24	178	116	1863

Tables 10 and 11 present the adjusted residual scores, highlighting significant behavioral patterns related to the types of feedback and their specificity. These z-scores were calculated to determine the statistical significance of event sequences. In particular, a value greater than + 1.96 is considered significant, with each significant entry marked by a single asterisk (*). Figure 5 graphically represents the significant transitions in learning behaviors for both the HG and LG learners. The arrows in the figure indicate the sequences of actions undertaken by learners, while the coefficients indicate the correlations among the eight identified events.

Comparing and contrasting the behavioral patterns between the two groups of learners in their collaboration with the AI virtual tutor revealed both similarities and differences (refer to Fig. 5). First, in the diagrams of both the HG and LG learners, a behavioral

Table 10 Adjusted residuals of the HG learners' FCM learning behaviors (z-scores)

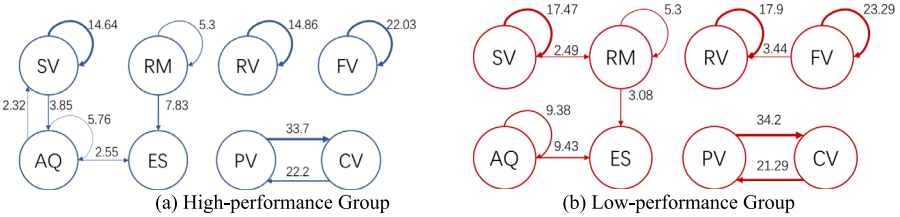
	SV	RV	FV	PV	CV	RM	AQ	ES
SV	14.64*	− 4.1	− 3.43	− 4.97	− 15.1	0.56	3.85*	0.33
RV	− 6.24	14.86*	− 0.89	1.29	1.59	− 1.05	− 0.9	− 3.09
FV	− 2.57	− 0.91	22.03*	− 3.25	− 4.16	− 1.83	− 3.34	− 0.94
PV	− 12.95	− 1.13	− 4.99	− 7.42	33.7*	0.79	− 2.08	− 0.69
CV	− 6.87	0.18	− 2.83	22.2*	− 5.7	− 1.62	− 2.62	− 2.8
RM	− 1.93	0.24	− 1.83	− 1.84	− 2.08	5.3*	− 0.22	7.83*
AQ	5.76*	− 2.88	− 2.26	− 4.59	− 4.03	− 0.22	2.32*	2.55*
ES	0	0	0	0	0	0	0	0

*z > 1.96

Table 11 Adjusted residuals of the LG learners' FCM learning behaviors (z-scores)

	SV	RV	FV	PV	CV	RM	AQ	ES
SV	17.47*	− 4.32	− 4.42	− 1.77	− 14.34	2.49*	0.24	− 0.4
RV	− 5.04	17.9*	1.16	0.02	− 2.6	− 1.21	− 1.38	− 2.3
FV	− 4.1	3.44*	23.29*	− 4.4	− 4.2	− 1.41	− 3.42	− 2.1
PV	− 12.6	− 3.26	− 5.73	− 10.75	34.2*	− 0.41	− 0.5	− 2.2
CV	− 4.11	− 2.34	− 2.15	21.29*	− 8.87	− 1.24	− 4.12	− 2.97
RM	1.05	− 1.21	− 0.63	− 2.46	− 2.31	3.08*	1.19	4.68*
AQ	0.92	− 2.06	− 3.72	− 5.42	− 4.53	− 0.9	9.38*	9.43*
ES	0	0	0	0	0	0	0	0

*z > 1.96



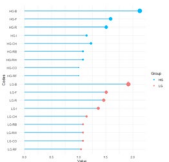
(a) High-performance Group (b) Low-performance Group

Fig. 5 Sequential-analysis transition diagrams for the HG and LG learners

Table 12 Frequency analysis results of the dialogue between learners and the AI virtual tutor

Dialogue form	Codes	HG	LG	Total
Building on ideas	B	124	72	196
Focusing	F	29	22	51
Reasoning	R	22	19	41
Inquiry	I	4	13	17
Challenge	CH	7	4	11
Reference back	RB	2	2	4
Reference to wider context	RW	2	2	4
Co-ordination	CO	0	2	2
Reflection	RF	0	1	1

Fig. 6 Lollipop chart illustrating frequency analysis results of dialogue between learners and the AI virtual tutor



Note: In order to enhance the rational layout of the images, this study subjected the data to a base-10 logarithmic transformation.

sequence AQ→ES is evident, indicating that learners tended to pose questions to the AI virtual tutor before concluding their learning sessions. However, there were no significant behavioral sequence relationships between RV, FV, PV, CV, and AQ, suggesting that learners rarely collaborated with the AI virtual tutor during the midpoint of video-based learning. Second, both the HG and LG diagrams exhibit the behavioral sequence AQ→AQ, indicating that learners repetitively asked questions and engaged in multi-turn dialogues with the AI virtual tutor. Third, unlike the LG learners, the behavioral patterns of the HG learners showed significant SV→AQ and AQ→SV sequences, suggesting that HG learners were more likely to interact with the AI virtual tutor by posing questions soon after the start of video learning and even before the commencement of video playback.

Collaborative knowledge building patterns from dialogue analysis

In the framework of the FCM, we scrutinized the collaborative dialogue forms between learners and the AI virtual tutor in knowledge building, as presented in Table 12 and Fig. 6. These elucidate descriptive statistical findings including 190 valid dialogue records for the HG learners and 137 records for the LG learners.

There is a notable resemblance in the knowledge-building dialogues between the two groups of learners. Within the dialogue with the AI virtual tutor, learners most frequently engaged in discourse centered around collaboratively building on ideas. Additionally, there was a tendency to guide the dialogue focus towards specific instructional contexts. However, only one dialogue from an LG learner was coded as reflective discourse (RF), suggesting a limited occurrence of reflective discourse in their collaborative knowledge-building dialogues.

The ENA results of the dialogue data between learners and the AI virtual tutor further illustrated distinctions in knowledge building between the HG and LG groups. A

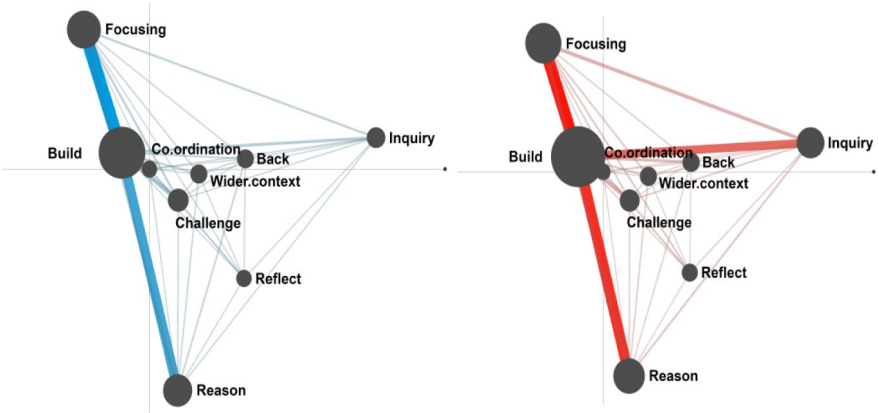
Mann–Whitney U test (Swiecki et al., 2020) was employed to compare the distribution of projection points in the ENA space between high and low academic performance. A statistically significant difference was observed at the alpha = 0.05 level on the X axis (Mdn = −0.01, $N = 190$, U = 15,974.00, $p = .00$, $r = -0.23$). This result indicates a statistically significant difference in the dialogue patterns between the HG and LG groups, suggesting that the HG learners exhibited a higher inclination towards a focusing-oriented approach, while the LG learners leaned more towards an inquiry-based direction.

The visualization employed network graphs, where nodes represented codes, and edges illustrated the relative frequency of code co-occurrence (refer to Fig. 7). Our model exhibited robust co-registration correlations of 0.96 (Pearson) across two dimensions, indicating a strong goodness of fit. The comparison of dialogue patterns between the two groups of learners revealed a pronounced collinearity between the behaviors of "Building on ideas" and "Focusing," as well as between "Building on ideas" and "Reasoning" (see Fig. 7). LG learners showed higher connection coefficients between "Building on ideas" and "Inquiry" (0.66) compared to the HG group (0.22), and higher coefficients for "Focusing - Inquiry" (0.28 for LG, 0.16 for HG).

Discussions

Impact on academic performance (RQ1)

The research findings provide empirical data to support the effectiveness of incorporating Generative AI and the Metaverse in FCM. Regarding the retention test, the virtual tutor in AIGM-FCM effectively mitigated cognitive overload and fully leveraged the benefits of FCM and virtual learning (Cheng et al., 2022b). With instant responses to their prompts, learners actively extended, questioned, and established connections between learning concepts, thereby generating a comprehensive understanding and improved retention test scores. When it comes to the transfer test, the technology-enhanced interaction and feedback facilitated by AIGM-FCM played a significant role (Ranieri et al., 2021). The innovative FCM tightly integrated knowledge acquisition and application, creating an



(a) High-performance Group (b) Low-performance Group

Fig. 7 The analyzed ENA networks of dialogue with the AI virtual tutor for the HG and LG learners

environment that provided timely feedback and encouraged learners to pose new questions and confront challenges. This allowed students to actively apply their conceptual knowledge to a variety of situations, which eventually helped them perform better on transfer exams.

According to the study conducted by Tomas et al. (2019), students with lower prior knowledge levels frequently require additional instruction from teachers in the context of flipped learning. To address the lack of personalized guidance in traditional FCM, this study suggests that the integration of AIGM technology into FCM results in more significant improvements (Hedges' $g = 0.775$) for learners with less prior knowledge when compared to their counterparts with higher levels of knowledge. The affordances offered by AIGM in revitalizing FCM serve as more effective scaffolding for learners with limited prior knowledge since AI virtual teachers can provide supplementary explanations and examples, as highlighted by Chernikova et al. (2020). The supportive role played by the AI virtual tutor in FCM was also evident through dialogue analysis in the process of collaborative knowledge building among learners.

Characteristics of learning behavior sequence (RQ2)

Questioning timing

The results of LSA showed a divergence in questioning patterns between HG and LG learners in their interactions with the AI virtual tutor. HG learners tended to ask questions (AQ) both before and after the initiation of an instructional video (SV), whereas LG learners commenced their questioning only upon the completion of the entire learning session (ES). Compared to the results of other studies on video-based learning and questioning behaviors, the findings in this study could be attributed to disparities in learners' prior knowledge and self-regulation levels (Dogan & Yucel-Toy, 2022; Li, 2019). Li's (2019) investigation suggested that learners possessing more extensive domain knowledge and a more comprehensive conceptual framework require less time to comprehend video learning materials. They engage with only a small fraction of the video, subsequently executing operations to monitor their progress and grasp the material. Simultaneously, posing questions to the AI virtual tutor serves as a self-regulation strategy, particularly for those students who are aware of their learning deficiencies and aspire to rectify these shortcomings (Dogan & Yucel-Toy, 2022).

Furthermore, neither the HG nor LG learners exhibited evident behavior of pausing the instructional videos for questioning. This result could be ascribed to their reluctance to disrupt the flow experience of video viewing (Fidan & Debbag, 2023). In particular, when compared to the conversational agents embedded in video learning platforms in prior research, the AI virtual tutor in this study operated separately from the video viewing experience (Winkler et al., 2020). In other words, the connection between the AI virtual tutor and the video content was not as integrated, potentially diminishing learners' inclination to interrupt the video for questioning.

Continuous questioning behavior

Both HG and LG learners demonstrated a tendency to repetitively pose questions to the AI virtual tutor. This finding indicates that learners were taking up epistemic agency when supported by generative AI, thereby fostering their knowledge-building (KB) effort

(Scardamalia & Bereiter, 2022). This behavior can be understood in light of the capabilities inherent in generative artificial intelligence itself. On one hand, the AI virtual tutor, powered by a LLM, engaged learners in interactive sessions, demonstrating superior capabilities in multi-turn dialogues and contextual reasoning compared to other natural language processing agents (Cao et al., 2023). On the other hand, Lim and Zhang (2022) found that the perception of contingency significantly influences the adoption of artificial intelligence technologies. Unlike task-based or rule-based conversational agents, the AI virtual tutor autonomously generates responses by learning patterns from data; thereby positively impacting the multiple interactions learners engage in. Notably, LG learners manifested a more evident inclination toward repetitive questioning in contrast to HG learners, a trend also consistent with subsequent ENA results.

Dialogue analysis of knowledge building (RQ3)

The level of knowledge building was basic

According to the ENA results, both HG and LG learners primarily employed dialogue forms such as "Building on Ideas," "Focusing," and "Reasoning" when engaging in knowledge building with the AI virtual tutor. In this context, learners asked the tutor questions based on a specific idea, seeking additional information and explanations. This observed phenomenon aligns with previous research on dialogue agents that facilitate learners' reasoning and argumentation to enhance learning performance (Tegos et al., 2015). However, dialogues between learners and the AI virtual tutor in FCM shown limited higher-order thinking, such as collaboration and reflection. This scarcity may be attributed to the prevailing interaction style, where dialogue agents often dominate conversations with learners. Consequently, learners have yet to develop the knowledge and skills required for initiating collaboration with the AI virtual tutor, thus keeping human-AI collaboration at a basic level (Kuhail et al., 2023). This observation is consistent with the findings of Chen et al. (2023) who explored students' knowledge building through direct interactions with ChatGPT. Their study highlighted learners' fundamental understandings of ChatGPT and the crucial role of human teacher guidance. Moreover, the inherent design of generative AI lacks sufficient accuracy and pedagogical capabilities, such as responding to queries with Socratic style of questioning. The gap between artificial intelligence and human intelligence may further inhibit achieving the desired level of collaborative knowledge building between learners and AI tutors and necessitate the integration of human oversights.

Learners tend to focus knowledge building on practical teaching scenarios

"Focusing" was the second most prevalent and the most frequently co-occurring dialogue form with "Building on ideas." When discussing the results of ENA, it is imperative to revisit the original data to comprehend the response patterns, a process referred to by Shaffer (2017) as "closing the interpretive loop." For instance, after posing the question "What is the future of artificial intelligence?" one student further narrowed the focus to "Are there any case studies of AI applications in education?" 'The shift towards a "Focusing" discourse may stem from the participant's practical teaching experience, emphasizing a concentration on how technological interventions can enhance professional practices in education (Teo, 2015). Concurrently, this outcome also underscores the necessity of training

domain-specific generative artificial intelligence to provide professional assistance (Pal et al., 2023).

Low-grade learners made more inquiries to the AI virtual tutor

Regarding inquiry-based discourse with the AI virtual tutor, LG learners were more likely than HG learners to engage, whether it was in the co-occurrence of "Building on ideas" and "Inquiry" or "Focusing" and "Inquiry." The primary reason for this phenomenon could be that the AI virtual tutor offered extra support to students who were struggling with flipped learning (Lee & Choi, 2019). Nonetheless, it is noteworthy that the combined findings of the ENA and LSA revealed that LG students were more receptive to asking the AI virtual tutor questions. The ease with which responses were obtained in this interaction raises the possibility that LG learners could become overly dependent on the AI virtual tutor, potentially undermining their capacity for critical thinking and problem solving (Kasneci et al., 2023).

Theoretical and practical implications

Theoretical implications

Firstly, this study expands on the FCM by integrating an AI virtual instructor within the context of the emerging generation of generative AI and the Metaverse. Enhancing pre-class learning effectiveness in the FCM has been a recurring challenge (Akçayır & Akçayır, 2018). By leveraging generative AI and the Metaverse, this study introduces a virtual instructor to provide learners with advanced assistance and guidance in pre-class learning, thus proving the viability of the strategy and advancing the theoretical framework and research on FCM. The integration of AIGM enhances the FCM from a constructivist perspective by emphasizing learners' active interactions with the AI virtual tutor, thereby better supporting students in actively engaging with the learning material rather than passively receiving information (Bognar et al., 2019; Lo & Hew, 2023). It addresses the need to build applications of AI in theory-based use in the classrooms (Díaz & Nussbaum, 2024).

Secondly, this research broadens the inquiry space of educational dialogue. Beyond traditional classrooms, technology-mediated teaching contexts and the expanded space of the internet are increasingly recognized as venues for instructional discourse (Wegerif & Major, 2019). Through incorporating the AIGM framework into FCM, this study enlarges the multimodal dialogue space for between learners and generative AI, and learners' active construction of knowledge is enhanced (Sanchez-Gonzalez & Terrell, 2023). Although the current state of human-AI collaborative knowledge building remains at a basic level, the interaction between learners and the AI virtual tutor advances social constructivist learning by harmonizing with pedagogical strategies that prioritize student-centered knowledge construction (Zhou & Schofield, 2024).

Finally, by integrating LSA and ENA, this study carefully examines the process of human-AI collaborative knowledge building between higher education students and the AI virtual tutor. This approach demonstrates how ENA and LSA can complement each other effectively, with ENA exploring the discourse content of dialogue and LSA situating learner-AI interactions within the entire flipped learning behavioral sequence. In a technology-mediated environment, this study reveals the interactional accomplishment of

intersubjective meaning-making between learners and AI to some extent, contributing to Scardamalia and Bereiter's (2022) theory on using technology and communities to support progressive knowledge-creating discourses.

Practical implications

Based on the results of this study, it is suggested that instructional design should prioritize tailored learner situations and adhere to learner-centered principles when integrating generative AI and the Metaverse into higher education. Firstly, the effective use of an AI virtual tutor may be more beneficial for disadvantaged learners than for those with exceptional learning performance, assisting those who are falling behind to catch up. Secondly, as generative artificial intelligence becomes increasingly prevalent in higher education, it is essential to enhance students' higher-order cognitive abilities, including their critical thinking, self-regulation, and AI literacy (Roffi et al., 2023). For instance, Wang et al. (2023a) suggest that students can interact more productively with AI virtual tutors within the FCM if they possess essential AI literacy and can master more effective prompts. Thirdly, although students' overall performance has improved, findings from the ENA indicating the limited competence of generative AI in supporting learners' collaborative knowledge building, coupled with the inherent inaccuracies and biases of LLMs, collectively underscore the importance of integrating human oversight and instruction in interactions between learners and generative AI in FCM (Baidoo-Anu & Ansah, 2023).

In terms of technological integration, generative AI and the Metaverse are expected to meet the demands of higher education closely. To better address the diverse needs of students and teachers, domain-specific generative artificial intelligence trained on educational corpus is necessary (Kasneci et al., 2023). Furthermore, generative AI is envisioned to directly evaluate different educational materials in the Metaverse learning environment and provide students with accurate, real-time feedback. For example, Suh and An (2022) developed a visual programming environment where student inputs in the code editing section were automatically analyzed by the generative AI model GPT-3 to produce images, aiding in code comprehension and promoting computational thinking.

Limitations and future directions

Despite implementing a strict process, this study still faces several limitations. Firstly, it did not employ a randomized controlled trial, so indicating a need for more empirical research evidence. Secondly, due to limitations in sample size, we employed the median split method for grouping. However, categorizing students based on their post-test scores using a median split, which ignores the continuum of their performance, can introduce bias and affect the interpretation of the behavioral and dialogue differences observed. More effective grouping methods such as clustering algorithms could be considered. Thirdly, the participants were all enrolled in the same course at the same university, suggesting future research should consider a broader diversity of learners and teaching. Fourthly, the LSA used in this study experienced delays and could not intervene in students' learning behavior promptly.

For future research directions, with the continuous advancement of LLM development frameworks like LangChain, research should delve deeper into the application of domain-specific generative artificial intelligence in higher education. Also, it is essential for generative AI to prioritize diverse training data and implement bias-mitigation algorithms

to update and improve LLMs, thus guaranteeing fair treatment for all students (Ferrara, 2023). Additionally, the emergence of multi-agent systems offers prospects for collaborative human-AI learning involving multiple agents. In terms of instructional design, incorporating additional knowledge-building principles, such as "Community knowledge" and "Constructive uses of authoritative sources," could enhance collaborative learning between learners and AI virtual tutors (Scardamalia & Bereiter, 2022).

Conclusions

In response to the significant impact that generative AI and the Metaverse have had on higher education, along with the ensuing concerns and discussions, this study investigated the efficacy of AIGM-FCM in higher education. The goal was to uncover the underlying mechanisms through careful instructional design, rigorous experimental procedures, and comprehensive data analysis. The findings reveal that integrating FCM with generative AI and the Metaverse can enhance students' learning outcomes in higher education. Throughout this process, learners and the AI virtual tutor engage in distinct patterns of interaction, such as posing questions at the beginning or end of videos and asking follow-up questions. Additionally, the results of ENA suggest that current interactions and knowledge-building with AI virtual tutors among higher education students are still at a very basic level. While most students attempted to maintain focused conversations, those in the lower scoring group were more inclined to use the AI virtual tutor for their inquiries compared to their higher-performing peers. These findings not only provide empirical evidence on the effectiveness of AIGM-FCM in higher education, but also offer valuable insights for practical instructional design and the integration of advanced technology. Future research in this area can build upon these insights and address the identified challenges.

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Data availability The data that support the findings of this study are available upon reasonable request.

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